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Bone tumor surgery device

Abstract

A bone tumor surgery device according to an embodiment of the present invention comprises: an accommodation container in which an accommodation space for accommodating a bone of a patient is formed; and a radiofrequency supply unit for supplying radiofrequency into the accommodation space.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a national phase application under 35 U.S.C. § 371 of International Application No. PCT/KR2020/017871, filed Dec. 8, 2020, which claims priority to and the benefit of Korean Application No. 10-2019-0163723, filed Dec. 10, 2019. The contents of the referenced patent applications are incorporated into the present application by reference.

TECHNICAL FIELD

(2) The present invention relates to a bone tumor surgery device, and more particularly, to a bone tumor surgery device that can be applied to a limb salvage operation among surgical treatment of musculoskeletal tumors.

BACKGROUND

(3) From several decades to the present, wide resection and reconstruction have been considered the standard of limb salvage operation in the surgical treatment of musculoskeletal tumors. That is, the methods of reconstructing the bone defect through tumor-prosthesis, allograft, and recycled autograft after wide resection of the patient's bone where the tumor exists is widely used. However, this method is pointed out as a problem in bone loss due to wide resection of the lesion, and the possibility of graft fracture and infection, etc.

(4) Meanwhile, pedicle frozen autograft surgery is sometimes used to overcome the disadvantages of recycled autograft among the above methods. The pedicle frozen autograft surgery is performed in a way that after performing osteotomy only in the metaphysis of the long tubular bone with a relatively good union rate, the tumor outside the bone and inside the medullary cavity is removed, then the part containing the tumor is moved out from the surgical field, and then freeze and thaw treatment is repeated.

(5) However, in the pedicle frozen autograft surgery, there are problems such as a possibility of contamination of the surgical area by tumor cells during the removal of tumors outside the bone and inside the medullary cavity; the risk of the operator having to hold the limb for a long time after by oneself lifting the limb and putting it in the liquid nitrogen container; and the non-uniformity of the bone cooling temperature.

(6) In such a situation, in order to minimize bone resection and preserve the patient's bones as much as possible, a tumor surgery technique using microwave is also being introduced. This is based on the fact that tumor cells die at a certain temperature range.

(7) However, conventional surgical devices using microwave employ catheters, coaxial antennas, etc., it is practically difficult to apply to a wide range of tumor removal because microwave is irradiated limitedly to the patient's bone by using such devices, and even if multiple devices are used, it is impossible to determine the device insertion density for uniform temperature transfer. In addition, it is impossible to safely collect live tumor cells from within the surgical field prior to insertion of the microwave surgical device, making pathological determination of the tumor impossible.

(8) Accordingly, it is required for the development of new technologies for limb salvage operation through which stable operation is possible through application within the surgical field, live tumors can be safely removed and harvested without the possibility of tumor contamination, and complete tumor death is possible by consistently delivering the targeted temperature.

SUMMARY OF THE INVENTION

Technical Problem

(9) The present invention is directed to providing a bone tumor surgery device capable of stably

removing a tumor without the possibility of contamination in a surgical field without wide bone resection.

(10) In addition, the present invention is directed to providing a bone tumor surgery device that effectively transfers heat for killing bone tumors to a patient's bone.

Technical Solution

(11) In an aspect, the present invention provides a bone tumor surgery device comprising an accommodation container in which an accommodation space for accommodating a bone of a patient is formed; and a radiofrequency supply unit for supplying radiofrequency into the accommodation space.

(12) In this case, the radiofrequency supply unit may include a radiofrequency generator; a heating electrode coupled to the accommodation container for transmitting the radiofrequency generated by the radiofrequency generator into the accommodation space; and a cable connecting the radiofrequency generator and the heating electrode.

(13) In addition, the cable may be a coaxial cable including a center conductor, and an outer conductor insulated from the center conductor and disposed to surround the center conductor; the center conductor may be connected to the heating electrode; and the outer conductor may be connected to the accommodation container.

(14) In addition, the bone tumor surgery device may further include a cooling fluid supply unit for supplying cooling fluid into the accommodation space.

(15) In addition, the cooling fluid supply unit may include an inlet conduit for supplying the cooling fluid and a supply conduit connected to the inlet conduit and disposed in the accommodation space.

(16) In addition, the supply conduit may have one or more outlets formed along its longitudinal direction so that the cooling fluid is discharged into the accommodation.

(17) In addition, the accommodation space may be formed in a cylindrical shape.

(18) In addition, the accommodation container may include a lower container forming a lower portion of the accommodation space and an upper container coupled to the lower container and forming an upper portion of the accommodation space.

(19) In addition, the accommodation container may include a window formed to allow the accommodation space to be observed from the outside.

(20) In addition, the accommodation container may include an insertion hole formed through one side thereof to insert the bone into the accommodation space.

(21) According to the bone tumor surgery device according to an embodiment of the present invention, it is possible to efficiently and completely remove a bone tumor by irradiating radiofrequency to the patient's bone disposed in the accommodation space of the accommodation container.

(22) In addition, according to the bone tumor surgery device according to an embodiment of the present invention, through efficient heat treatment using radiofrequency, surgical treatment of bone tumors can be achieved without wide bone resection, reconstruction using tumor-prosthesis or allograft.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects, features, and advantages of the present invention will become more apparent to those of ordinary skill in the art by describing embodiments thereof in detail with reference to the accompanying drawings, in which:

(2) FIG. 1 is a view showing a bone tumor surgery device according to an exemplary embodiment of the present invention;

- (3) FIG. 2 is a view showing a state of usage of the bone tumor surgery device according to an exemplary embodiment of the present invention;
- (4) FIG. 3 is an exploded view of an accommodation container part of the bone tumor surgery device according to an exemplary embodiment of the present invention;
- (5) FIG. 4 is a view showing the accommodation container into which the patient's bones are inserted, taken along section A-A' of FIG. 1.

DETAILED DESCRIPTION OF THE EMBODIMENTS

- (6) Hereinafter, exemplary embodiments of the present invention will be described in detail so that those of ordinary skill in the art can readily implement the present invention with reference to the accompanying drawings. The present invention may be embodied in many different forms and are not limited to the embodiments set forth herein. In the drawings, parts unrelated to the description are omitted for clarity. Throughout the specification, like reference numerals denote like elements.
- (7) It is understood that the terms “comprise” or “have” when used in this specification, are intended to describe the presence of stated features, integers, steps, operations, members, components and/or a combination thereof but not preclude the possibility of the presence or addition of one or more other features, integers, steps, operations, members, components, or a combination thereof.
- (8) In addition, in this specification, spatially relative terms “front”, “rear”, “upper or above or top” or “lower or below or bottom” may be used to describe a correlation with the elements shown in the drawings. These are relative terms determined based on what is shown in the drawings, and the positional relationship may be conversely interpreted according to the orientation.
- (9) In addition, the presence of an element in/on “front”, “behind”, “above” or “below” of another element includes not only being disposed in/on “front”, “rear”, “above” or “below” directly in contact with other elements, but also cases in which another element being disposed in the middle, unless otherwise specified.
- (10) FIG. 1 is a view showing a bone tumor surgery device according to an exemplary embodiment of the present invention; FIG. 2 is a view showing a state of usage of a bone tumor surgery device according to an exemplary embodiment of the present invention. In addition, FIG. 3 is an exploded view of an accommodation container part of the bone tumor surgery device according to an exemplary embodiment of the present invention. FIG. 4 is a view showing the accommodation container into which the patient's bones are inserted, taken along section A-A' of FIG. 1.
- (11) A bone tumor surgery device according to an embodiment of the present invention allows to kill the tumor T using radiofrequency in a state that accommodates the bone B of the patient P in the accommodation container **10**, and to reuse the patient's own bone B as much as possible. Specifically, as shown in FIG. 2, in case the bone tumor surgery device according to an embodiment of the present invention is applied to long tubular bone containing a tumor, it is possible to kill tumor cells in situ through radiofrequency after resection of the metaphysis or epiphysis of the long tubular bone (the other side of the long tubular bone is not resected). According to an embodiment of the present invention, a patient can maintain a highly durable limb for a long time through a simple operation.
- (12) Referring to FIGS. 1 to 4, the bone tumor surgery device according to an embodiment of the present invention may include an accommodation container **10**, a radiofrequency supply unit **20**, and a cooling fluid supply unit **30**.
- (13) The accommodation container **10** provides an accommodating space **11** for accommodating the bone B of the patient P. The accommodation space **11** is a portion for accommodating the bone B of the patient P in the accommodation container **10**. According to an embodiment of the present invention, the accommodating space **11** is formed in the accommodating container **10** in a state in which a lower container **12** and an upper container **13** are coupled. A radiofrequency (RF) is transmitted to the bone B of the patient P disposed in the accommodation space (**11**), and heat treatment for the death of the tumor cells T is performed.

- (14) The accommodation container **10** is preferably made of a material or structure having dielectric properties that can block the radiofrequency in the accommodation space **11** from leaking to the outside. That is, it is preferable that the material of the accommodation container **10** can be insulated and shielded. In addition, the accommodation container **10** may be made of a material that can be sterilized.
- (15) As an example of a material that satisfies the above characteristics, a metal plate, a metal mesh structure, or a plastic material or the like may be used. More specifically, stainless steel, aluminum, or the like may be used.
- (16) As best shown in FIG. **3**, in an embodiment of the present invention, the accommodation container **10** may include a lower container **12** forming a lower portion of the accommodation space **11** and an upper container **13** coupled to the lower container **12** and forming an upper portion of the accommodation space **11**. More specifically, the lower container **12** and the upper container **13** may form a cylindrical shape in the coupled state and form a cylindrical accommodation space **11** therein.
- (17) The lower container **12** is formed in a semi-cylindrical shape with an open-top. A first protrusion **121** is formed in at least a portion of the lower container **12** in contact with the upper container **13**. The first protrusion **121** is a portion to which a fastening part **16** to be described later is coupled for stable coupling of the lower container **12** and the upper container **13**. The first protrusion **121** may protrude outward by a predetermined length from the outer surface of the lower container **12**, and the outer end thereof may additionally protrude downward so that the transverse section thereof may have a shape such as ‘ $\neg$ ’.
- (18) The upper container **13** is formed with an open bottom so as to cover the open portion of the lower container **12**. That is, the upper container **13** may have a cover shape that covers the open portion of the lower container **12**.
- (19) A second protrusion **131** is formed in a portion of the upper container **13** in contact with the first protrusion **121** of the lower container **12**. The second protrusion **131** is a portion to which a fastening part **16** to be described later is coupled for stable coupling of the lower container **12** and the upper container **13**. Specifically, the second protrusion **131** may protrude outward by a predetermined length from the outer surface of the upper container **13**, and the outer end thereof may additionally protrude upward so that the transverse section thereof may have a shape such as ‘ $\lceil$ ’.
- (20) Further, in an embodiment of the present invention, the accommodation container **10** includes a window **14** that allows the accommodation space **11** to be observed from the outside. Specifically, according to an embodiment of the present invention, the window **14** is formed on the upper container **13**. The surgeon performing the operation may visually check the bone B of the patient P disposed in the accommodation space **11** through the window **14**.
- (21) In addition, an insertion hole **15** for inserting the bone B of the patient P into the accommodation space **11** may be formed on one side of the accommodation space **11** of the accommodation container **10**. In this case, a sealing member (not shown) made of a material considering the protection of the bone B and prevention of radiofrequency leakage may be coupled to the insertion hole **15** in contact with the bone B of the patient P. For example, the sealing member may be a ring-shaped member made of a soft material to be fitted into the insertion hole.
- (22) In an embodiment of the present invention, the lower container **12** and the upper container **13** form an insertion hole **15** in the coupled state. To this end, the lower container **12** includes a first invagination portion **151** invaginated downward into one wall in the longitudinal direction, and the upper container **13** includes a second invagination portion **152** invaginated upward into a portion of one wall in the longitudinal direction in contact with the first invagination portion **151** of the lower container **12**. When the lower container **12** and the upper container **13** are coupled, the first invagination portion **151** and the second invagination portion **152** form one insertion hole **15**. The combination of the lower container **12** and the upper container **13** can be made in a state in which

the patient's bone B is placed in the accommodation space **11** of the accommodation container **10** through the insertion hole **15** formed in this way.

(23) The fastening part **16** fastens the coupling portion of the lower container **12** and the upper container **13**. Through the fastening part **16**, the lower container **12** and the upper container **13** can maintain a stable fastening state.

(24) As described above, in an embodiment of the present invention, the lower container **12** and the upper container **13** include the first protrusion **121** and the second protrusion **131**, respectively. In this regard, the fastening part **16** may be a length member (e.g., a sleeve) that surrounds and couples the first protrusion **121** and the second protrusion **131** together.

(25) More specifically, in an embodiment of the present invention, in a state in which the lower container **12** is covered by the upper container **13**, the first protrusion **121** and the second protrusion **131** come into contact and protrude outward by a predetermined length from the outer surface of the housing **10**, the outer end thereof is additionally protruded upward and downward so that its transverse section appears as '└┐'. Accordingly, the fastening part **16** may be made of a length member that is fitted while surrounding the abutting first protrusion **121** and second protrusion **131** and whose longitudinal cross-section appears in a shape complementary to '└┐'.

(26) In addition, the fastening part **16** may include a metal slit for preventing the radiofrequency and heat from leaking from the accommodation space **11**. More specifically, when the fastening part **16** is formed of a length member, a plurality of metal slits may be formed along the longitudinal direction of the fastening part **16**.

(27) The radiofrequency supply unit **20** supplies a radiofrequency into the accommodation space **11**. In an embodiment of the present invention, the radiofrequency supply unit **20** may include a radiofrequency generator **21**, a heating electrode **22** and a cable **23**.

(28) The radiofrequency generator **21** is a device for generating a radiofrequency that can heat the bone B of the patient P disposed in the accommodation space **11**. The radiofrequency generator **21** may receive power to generate a radiofrequency, match the generated radiofrequency and output it. The radiofrequency output from the radiofrequency generator **21** may be, for example, 25 to 30 MHz. More specifically, the radiofrequency may be 27.12 MHz. Of course, the range of the radiofrequency output from the radiofrequency generator **21** is not limited thereby, and any radiofrequency capable of heating the deep part of the bone B of the patient P to an appropriate temperature may be selected.

(29) In addition, the radiofrequency generator **21** may be configured to generate a radiofrequency in a specific range, but configured to be capable of setting power, time, and the like differently as needed. To this end, the radiofrequency generator **21** may provide an interface (e.g., a button, a knob, a touch-type display, etc.) capable of setting power, time, and the like.

(30) The heating electrode **22** transmits a radiofrequency into the accommodation space **11**. In an embodiment of the present invention, the heating electrode **22** includes an electrode body **221** and a connector **222**. The electrode body **221** is disposed in the accommodation space **11**. The electrode body **221** may be formed of a plate material. For example, the electrode body **221** may be curved in an arc shape. In addition, the connector **222** may be electrically connected to the electrode body **221** so that the electrode body **221** can be connected to the radiofrequency generator **21**, and may be provided outside the accommodation container **10**.

(31) The electrode body **221** may be disposed in an upper portion of the accommodation space **11**. Specifically, in an embodiment of the present invention, the electrode body **221** may be coupled to the upper container **13**. In addition, the electrode body **221** may be a first electrode (e.g., anode), and the lower portion of the container **10** may be a second electrode (e.g., cathode). Accordingly, the radiofrequency may be supplied to the bone B disposed between the electrode body **221** and the inner wall of the accommodation container **10** in the accommodation space **11**.

(32) The cable **23** connects the radiofrequency generator **21** and the heating electrode **22**. In an embodiment of the present invention, the cable **23** may be a coaxial cable including a center

conductor, and an outer conductor surrounding the center conductor in a state insulated from the center conductor by an insulator.

(33) In this case, the center conductor of the cable **23** may be connected to the electrode body **221** and the outer conductor may be connected to the accommodation container **10**, so that, as described above, a high frequency may be supplied to the bone B disposed between the electrode body **221** and the inner wall of the accommodation container **10** in the accommodation space **11**.

(34) The cooling fluid supply unit **30** supplies cooling fluid into the accommodation space **11**. When the radiofrequency is transmitted to the bone B of the patient P disposed in the accommodation space **11**, if the surface of the bone B is excessively heated, not only the tumor cells T but also normal cells may be damaged; the cooling fluid flows into the accommodation space **11** and acts to cool, thereby preventing local heating of the surface of the bone B. Through this, while avoiding excessive heating of the surface of the bone B through the radiofrequency, the radiofrequency may be sufficiently transmitted to the deep part of the bone B, so that it is possible to sufficiently heat the tumor cells in the deep part.

(35) For example, the cooling fluid supplied by the cooling fluid supply unit **30** may be air. That is, in an embodiment of the present invention, the cooling fluid supply unit **30** may supply air into the accommodation space **11**.

(36) In an embodiment of the present invention, the cooling fluid supply unit **30** may include a pneumatic regulator **31** connected to an external air tank (not shown) to supply air at a constant pressure, an inlet conduit **32** connected between the pneumatic regulator **31** and the accommodation space **11**, a supply conduit **33** connected to the inlet conduit **32** and disposed in the accommodation space **11**, and an outlet conduit **34** connected to the outside in the accommodation space **11**.

(37) In this regard, the accommodation container **10** may include a cooling fluid inlet **17** formed through one side of the accommodation space **11** and a cooling fluid outlet **18** formed through the other side of the accommodation space **11**. That is, the inlet conduit **32** may be connected to the cooling fluid inlet **17**, and the outlet conduit **34** may be connected to the cooling fluid outlet **18**.

(38) In addition, the supply conduit **33** may have one or more outlets formed along its longitudinal direction so that the cooling fluid is discharged into the accommodation space **11**. For example, the outlet may be formed of a through hole, a slit, or the like.

(39) In an embodiment of the present invention, the supply conduit **33** communicates with the inlet conduit **32** and is disposed in the accommodation space **11**, and by providing a plurality of outlets **331** in the form of discharge holes along the longitudinal direction thereof, allows the cooling fluid to be uniformly supplied into the accommodation space **11**.

(40) Although exemplary embodiments of the present disclosure have been described, the spirit of the present disclosure is not limited to the embodiments set forth herein. Those of ordinary skill in the art who understand the spirit of the present disclosure may easily propose other embodiments through supplement, change, removal, addition, etc. of elements within the same spirit, but the embodiments will be also within the scope of the present disclosure.

Claims

1. A bone tumor surgery device comprising: an accommodation container in which an accommodation space for accommodating the bone of a patient is formed, the bone being exposed outside the patient's skin; and a radiofrequency supply unit for supplying radiofrequency into the accommodation space, wherein the accommodation container comprises an insertion hole formed through one side thereof to insert the bone into the accommodation space, wherein the radiofrequency supply unit comprises a radiofrequency generator and a heating electrode coupled to the accommodation container for transmitting the radiofrequency generated by the radiofrequency generator into the accommodation space, and wherein the heating electrode

comprises an electrode body configured to be disposed in the accommodation space, the electrode body transmitting the radiofrequency to the bone disposed between the electrode body and an inner wall of the accommodation container.

2. The bone tumor surgery device of claim 1, wherein the radiofrequency supply unit further comprises a cable connecting the radiofrequency generator and the heating electrode.

3. The bone tumor surgery device of claim 2, wherein the cable is a coaxial cable including a center conductor, and an outer conductor insulated from the center conductor and disposed to surround the center conductor; the center conductor is connected to the heating electrode; and the outer conductor is connected to the accommodation container.

4. The bone tumor surgery device of claim 1, further comprising a cooling fluid supply unit for supplying cooling fluid into the accommodation space.

5. The bone tumor surgery device of claim 4, wherein the cooling fluid supply unit comprises an inlet conduit for supplying the cooling fluid and a supply conduit connected to the inlet conduit and disposed in the accommodation space.

6. The bone tumor surgery device of claim 5, wherein the supply conduit has one or more outlets formed along its longitudinal direction so that the cooling fluid is discharged into the accommodation space.

7. The bone tumor surgery device of claim 1, wherein the accommodation space is formed in a cylindrical shape.

8. The bone tumor surgery device of claim 7, wherein the accommodation container comprises a lower container forming a lower portion of the accommodation space and an upper container coupled to the lower container and forming an upper portion of the accommodation space.

9. The bone tumor surgery device of claim 1, wherein the accommodation container comprises a window formed to allow the accommodation space to be observed from the outside.
